Reflective Report on Database Project

This project focused on the design and implementation of a transportation management database system. Throughout the process, our team worked with several core database concepts, including entity-relationship modeling, relational schema design, SQL querying, and advanced database mechanisms such as triggers and functions. The experience provided both practical technical skills and a deeper understanding of how theoretical database concepts are applied in real-world systems.

One of the first key learnings was the importance of **Entity-Relationship (ER) modeling**. Designing the ER diagram required us to carefully analyze the system requirements and identify entities such as *Passenger, Ticket, Trip, Route, Station, Driver,* and *Vehicle*, along with their relationships. We learned that small design decisions at this stage significantly impact the entire system. For example, correctly modeling relationships like “Passenger buys Ticket” or “Trip uses Vehicle” helped avoid redundancy and ensured data consistency later in the relational schema.

Transforming the ER diagram into a **relational database schema (DDL)** was another critical step. We gained experience in defining primary keys, foreign keys, constraints, and normalization principles. Ensuring referential integrity using foreign key constraints (e.g., linking *Ticket* to *Passenger* and *Trip*) helped maintain consistency across tables. We also learned how constraints such as CHECK, NOT NULL, and UNIQUE contribute to enforcing business rules directly at the database level.

A major part of the project involved writing **SQL queries for data analysis and reporting**. Through this, we improved our ability to use joins, aggregations, subqueries, and grouping. For example, queries such as revenue analysis by ticket type, identifying frequent passengers, or detecting delayed trips required combining multiple tables and applying aggregate functions. This helped us understand how databases are not just for storage, but also powerful tools for extracting meaningful insights.

Another key area of learning was the use of indexes for performance optimization. By adding indexes to frequently accessed columns such as `ticket_id`, `payment_time`, and `trip_id`, we improved query execution speed and gained insight into how databases handle large volumes of data efficiently. This also helped us understand the trade-offs involved, as indexing can increase storage usage and slightly slow down insert or update operations.

The most technically challenging aspect of the project was implementing triggers and stored functions. These features enabled us to enforce complex business rules directly within the database. For instance, we implemented mechanisms to prevent scheduling conflicts for drivers and vehicles, ensure that payment amounts exactly match ticket prices, automatically update passenger counts after boarding, and validate route consistency by checking timing and duplicate stops. Working with these tools demonstrated how databases can go beyond simple data storage to actively maintain data integrity and enforce logic. It also emphasized the need to carefully consider edge cases and ensure that all constraints function correctly under different scenarios.

From a teamwork perspective, the project emphasized **collaboration and division of responsibilities**. Tasks such as ER design, DDL creation, query writing, and trigger implementation were distributed among team members. This required clear communication to ensure consistency across different parts of the system. We also learned the importance of testing and validating each component before integration.

One of the challenges we faced was ensuring that all parts of the system worked together correctly. For example, incorrect foreign key relationships or trigger logic could easily break the system. Debugging these issues improved our problem-solving skills and taught us to approach database development systematically.

In conclusion, this project significantly strengthened our understanding of database systems. We moved beyond theoretical knowledge and gained hands-on experience with designing, implementing, and optimizing a complex relational database. The combination of SQL, constraints, indexing, and triggers demonstrated how databases can support both data integrity and business logic. Overall, the project was highly valuable in preparing us for real-world applications of data management systems.